

# **Sensing Moisture Content of the Soil**

Project report submitted in partial fulfillment of the requirement for the award of the  
Degree of **B.E.**

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**Osmania University, Hyderabad,**

**Approved by AICTE, Accredited by NBA and NAAC with  
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**CERTIFICATE**

This is to certify that the project report entitled **Sensing Moisture Content of the Soil** being submitted Srija Yadav, Sanskriti Agarwal, and Sharia Zainab in partial fulfillment for the award of the Degree of Bachelor of Engineering in Information Technology to the Osmania University is a record of bona fide work carried out by them under my guidance and supervision.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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## **ACKNOWLEDGEMENT**

We take this opportunity to convey our heartfelt gratitude to each and every one who has supported us in every way or the other during the course of our project.

From the very core of my heart, we would like to express our sincere gratitude to our internal guide G. Sreelatha for her supervisory guidance. We express our profound gratitude to project coordinator Mrs Ruquia Alam and Dr.A.Kanaka Durga, HOD of the Information Technology department. We are always grateful to our peers and friends who have always encouraged us and guided us whenever we needed assistance.



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**DECLARATION**

We declare that the work reported in the thesis entitled “**Sensing Moisture Content of the Soil**” is a record of the work done by us in the Department of Information Technology in Stanley College of Engineering and Technology for Women, Hyderabad. No part of the thesis is copied from books / journals / internet and wherever referred the same has been duly acknowledged in the text. The reported data is based on the project work done entirely by us and not copied from any other source.

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Empower Women; Impact the World Empowering girl students through professional education integrated with values and character to make an impact in the World.

*The Mission STLW, in pursuance of its vision:*

M1: Providing quality engineering education for girl students to make them competent and confident to succeed in professional practice and advanced learning.  
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- 12) Lifelong Learning:** Recognize the need for, and have the ability to engage in independent and life-long learning

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**PSO1: Skilled Professional:** Ability to apply technical skills and involve in the Creation, maintenance and use of Computer, Computer Networks and Computer Information Systems.

**PSO2: Research Capability:** Ability to pursue research with academic excellence and core competence skills.

## ABSTRACT

India is mainly an agricultural country. Agriculture is the most important occupation for most of the Indian families. It plays a vital role in the development of agricultural countries. In India, agriculture contributes about 16% of total GDP and 10% of total exports. Efficient irrigation management can improve yields, grain quality, conserve water and energy, and reduce nutrient leaching. One of the easiest and most effective ways to improve irrigation efficiency is to **implement soil sensor technology in irrigation scheduling.**

Today agriculture is embedded with advanced sensors like Temperature, humidity and moisture sensors that enable them to communicate to each other, analyze the data and also exchange data among them. There has always been a vital role of agriculture in countries Like India and issues concerning agriculture have become very common these days. Smart agriculture can act as a pillar to the development of the country. Because water is an important resource and not all communities around the world can afford to be liberal with their water needs; it has become important to use available water as efficiently as possible, especially in agriculture.

As water is main resource for Agriculture, Irrigation is one method to supply water but in some cases there will be lot of water wastage For the purpose of reducing the over watering of crops, an unattended ground moisture sensor can be implemented to **measure the current moisture level in the soil** surrounding the plants. This will allow a farmer to know when to water/stop watering his crop.

The Internet of Things (IoT) is transforming the agriculture industry thereby enabling farmers to contend with enormous challenges they face. In this proposed system we are using sensors like soil moisture sensors which senses the soil and based on soil moisture value land can get irrigated accordingly.

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# CHAPTER 1

## INTRODUCTION

### 1.1 Introduction

The project aim is to detect the dryness in soil using sensors and so that water can get provided to the plants accordingly. This project helps to maintain the plants quite easily. In this project we are detecting a sensor and need for Irrigation.

**Sensing Moisture Content of the Soil** will serve the following purposes-

As there is no un-planned usage of water, a lot of water is saved from being wasted.

The irrigation is done only when there is not enough moisture in the soil and the sensor decides when the water should be provided. This saves a lot of time for the farmers. This also gives much needed rest to the farmers, as they don't have to go and check the moisture of the soil manually.

This system is designed for the purpose of solving different problems related to the irrigation process so this system mainly targets farmers and even those people who have gardens in their land.

Here it aims to develop a system to provide water when it is required for the plants which helps in saving water and money. And also helps in minimal wastage of water. It allows reinvestment in new and improved technologies which ensure sustainable and responsible irrigation over time. It also allows controlling the amount of water delivered to the plants when it is needed.

## 1.2 Existing System

### 1.2.1 Limitations

- The continuous increasing demand of food requires the rapid improvement in food production technology. In a country like India, where the economy is mainly based on agriculture and the climatic conditions are isotropic, still we are not able to make full use of agricultural resources. The existing irrigation system is tedious, time consuming and very expensive in terms of water consumption and operation.
- The existing method and one of the oldest ways in agriculture is the manual method of parameter control. It requires a lot of work and time and is generally done with a pipe or a bucket.
- It has several limitations, leaching off of soil nutrients, erosion due to flooding, loss of water from plant surfaces through evaporation, water wastage which can result in water scarcity in drought areas and production of unhealthy crops. Also due to seepage in drains, wastage of water is caused.
- The existing system of manual irrigation is very inefficient in regard to solving these issues. In modern drip irrigation systems, the most significant advantage is that water is supplied near the root zone of the plants drip by drip due to which a large quantity of water is saved.
- At the present era, the farmers have been using irrigation techniques in India through manual control in which farmers irrigate the land at the regular intervals.
- This process sometimes consumes more water or sometimes the water reaches late due to which crops get dried. Water deficiency can be detrimental to plants before visible wilting occurs. Slowed growth rate, lighter weight fruit follows slight water deficiency.
- This problem can be perfectly rectified if we use Sensing Moisture Content of the Soil in which the irrigation will take place only when there will be acute requirement of water.
- The main reason is the lack of rains & scarcity of land reservoir water. The continuous extraction of water from earth is reducing the water level due to which a lot of land is coming slowly in the zones of un-irrigated land.

- Another very important reason for this is due to unplanned use of water due to which a significant amount of water goes to waste.

## **1.3 Proposed System**

### **1.3.1 Advantages**

- The proposed system will allow farmers to continuously monitor the moisture level in the field. When moisture goes below a certain level, sprinklers would be turned on manually, thus achieving optimal irrigation.
- It also improves the traditional irrigation system, enabling the irrigation system to have high efficiency and low water usage. Proper usage of the irrigation system is very necessary because it causes shortage of land reserved water due to lack of rain, spontaneous use of water as a result large amounts of water goes waste.
- This system saves labour cost and water up to 70%. The aim of implementing this project was to demonstrate the automatic detection of moisture content of the soil for plant irrigation which can be used to reduce water consumption and save time.
- The main advantage of this project is that it has faster execution when compared to manual execution of the process. It is simple, portable and provides high performance.
- It consumes less power
- Dryness can be easily detected in soil.
- Permits a non- expert to do the work of an expert.
- Improves productivity by increasing work output and improving efficiency.
- Saves time in accomplishing specific objectives.
- This system ensures that the plants do not endure from the strain or stress of less and over watering.
- Reduces maintenance costs.
- Cheaper to prepare and it greatly diminishes the need for manpower by a great value

- This system will be able to function even when the owner is unavailable for a small period of time, hence ensuring proper irrigation even in the absence of people. Also water will not be wasted during traversal.

### **1.4 Problem Statement**

Nowadays, despite being an agricultural country, the number of people who die of hunger is still quite high. Access to food seems to be difficult, as price and quantity of food is still beyond the capability of the lower middle class and lower class. Irrigation induced Crop failure is a major cause of crop loss every year, and in the age of water crises, this has been elevated to great levels. In order to keep up with increasing demand, farmers are required to increase crop efficiency, by rapidly advancing technologies. In order to handle Irrigation issues, this system has been devised and implemented. Usually, farmers need large scale manpower to irrigate large lands simultaneously. However, the Sensing Moisture Content of the Soil is a system that facilitates irrigation of lands, upon need.

#### **1.4.1 Objectives and Applications**

The objectives to consider are:

- Simplify the irrigation system by installing and designing the whole irrigation system.
- Save energy, which allows the application of sensing moisture content can be used in more other applications.
- Optimize water consumption.
- Automated system fully.
- Decrease the cost of operation.
- Make the system easy to use by farmers.

## 1.5 Non- Functional Requirements

- **Communication**

Sensor connected to the LPC-1769 Microcontroller to take all the data from prototype to the Windows System

- **Accuracy**

The sensor is to take readings of all the data and store it in a controller and at the same time the status of the irrigation can be updated to the farmer.

- **Performance**

The system must work in real time.

- **Operational**

This system should detect the moisture content of the soil and it is connected to the Windows system. so, users must download keil microvision 4 to control.

- **Cost**

The cost of this system must not be too expensive because we aim to decrease the work which means decrease the amount of money in the irrigation process and solve the main problem i.e reduce the water consumption at the same time.

## 1.6 Functional Requirements

This system works through different stages

- If the moisture sensor is dry and is below the set limit it indicates that the switch of the pump should be turned ON.
- If the moisture sensor is wet and is above the set limit it indicates that the switch of the pump should be turned OFF.
- When the moisture level rises above the set point, the farmer switches off the pump. The status of the moisture level will be displayed. It is designed for proper watering of the plants and handled remotely by farmers.
- When the soil goes dry, the pump can start watering.
- Sensors when inserted into soil continuously check the moisture and send the data to the Electric Board of YL 38 comparator.
- The Microcontroller LPC-1769 is used to control this System and Windows is connected to the controller.
- The sensors are to take values from surroundings and store it in a controller. According to this value the drip can be ON/OFF.



## **1.7 System Requirements**

We make analysis of the system requirement based on different factors which are

### **1. Easy to implement**

The materials required for this project must be easy to install to implement a successful project. In addition, materials should be easy to connect with each other to build this project and become more effective. Also, the materials of this project must be easy to replace in case of any damage.

### **2. Open source**

The controller used for this project is open-source, so the used hardware is reasonably priced and has free development software.

### **3. Strength**

The tools needed for this project must be strong to operate for a long period of time to achieve the desired success. In addition to achieving one of the important goals required to save money.

### **4. Quality**

The tools required to build this project must be of excellent quality to operate for a long time. Selecting quality materials is very important to avoid wasting money and to avoid the technical problems of these devices and disturbance of the process. Excellent quality is required for this project to achieve the desired success of this project.

### **5. Modifiability**

The material should be chosen based on its ease of modifiability, as it is common to come across designs and connections. Also, to be easy to replace or modify required in the future.

### **1.7.1 Software Requirements**

- Windows System(32 bit system processor or 64 bit system processor)
- Keil MicroVision4
- Flash Magic Tool

### **1.7.2 Hardware Requirements**

- LPC 1769 Microcontroller
- Power Supply
  - Comparator
  - LED
- Sensor - YL69 Sensor
- Connecting Wires

## **1.8 Overview of the Report**

In late decades, there is a quick advancement in Smart Agricultural Systems. Show that agriculture has great importance worldwide. Indeed, in India for example, about 70% of the people rely upon the vital sector of agriculture. In the past, irrigation systems used to be dependent on the mills to irrigate the farm by conventional methods without knowing the appropriate quantities of these crops. These old systems are a major cause of the waste of large quantities of water and thus destroy some crops because of the lack of adequate quantities of water. However, with the recent technological developments, there have been innovative systems for irrigation without the farmer interfering in the irrigation process.

Agriculture is the backbone of the Indian Economy. In today's world, as we see rapid growth in global population, agriculture becomes more important to meet the needs of the human race.

However, agriculture requires irrigation and with every year we have more water consumption than rainfall, it becomes critical for growers to find ways to conserve water while still achieving the highest yield. But in the present era, the farmers have been using irrigation techniques through the manual control in which they irrigate the land at the regular interval.

According to statistics, agriculture uses 85% of available freshwater resources worldwide, and this percentage will continue to be dominant in water consumption because of population growth and increased food demand. There is an urgent need to create strategies based on science and technology for sustainable use of water, including technical, agronomic, managerial and institutional improvements. Agricultural irrigation based on Internet technology is based on crop water requirement rules. By using Internet technology and sensor network technology we can control water wastage and to maximize the scientific technologies in irrigation methods. Hence it can greatly improve the utilization of water and can increase water productivity.

Agriculture is considered as the basis of life for us as it is the main source of food and other raw materials. It plays a vital role in the growth of a country's economy. Growth in the agricultural sector is necessary for the development of the economic condition of the country. Unfortunately, many farmers still use the traditional methods of farming.

In India most of the irrigation systems are manually operated one's. These outdated techniques are replaced with automated techniques, this focuses primarily on reducing the wastage of water and minimizing the manual labor on fields for irrigation.

## **CHAPTER 2**

# **LITERATURE SURVEY**

### **2.1 Patent Search**

Three patents regarding moisture sensors were found. The first one is US 5424649 A, published on June 13, 1995. The patent is for a soil moisture sensor using sensor electrodes to sense capacitance which is proportional to the moisture of the soil. It is similar to the proposed design because both the patented project and the idea regards sensing the moisture of soil. However, there are many differences between the project and the patented one. Primarily, the device will be self sufficient, in that it will not require being plugged into a wall. This device will also record the data measured and send it to the user for statistical analysis. The second patent is US 20080199359 A1, published on August 21, 2008. The patent is for a soil moisture sensor which measures the moisture of the soil and stores the values in its memory. It is similar to our proposed design because, like the previous patent, both the patented project and our idea regards sensing the moisture of soil. Both projects also store the sensor readings in the memory. However, this project is different because it will provide the stored data to the user for statistical analysis. The third patent is US 5621669 A, published on April 15, 1997. It is a probe used to measure moisture sensors and a controller for external actuators [3]. It is similar to our proposed design because, again, both the patented project and our idea regards sensing the moisture of soil. However, it is different because this device is simply a probe, not a fully-functioning measurement system. It does not store the measured values, nor does it provide the user with the data. Finally, it can be used as a controller, for which this project cannot be used.

## **2.2 Energy Harvesting Component**

Due to the fact that this moisture sensor system needs to be as automated and as self-sufficient as possible, an energy harvesting component needs to be added to the system. This can be done numerous ways, however, since this system is designed to be a low-cost system that can be used in various areas, a solar panel is the best option for this system. The solar panel will power all of the electronics in the soil moisture sensor. However, solar panels experience many issues in supplying a constant power, especially on cloudy days. Thus, a backup battery will be needed for the solar panel. On sunny days, the solar panel will power the whole system and will charge the backup battery supply. The battery will power the electronics when the solar panel cannot. For safe measure, a sensor should be attached to the solar panel and the DC battery supply to measure the power output. Due to weather conditions, there is a chance that the power output could be zero because no energy has been harvested over a long period of time. If this were to happen, a notification would be sent to the user informing him/her to switch the depleted battery with a fully charged battery. In the article, “Automated Irrigation System Using Solar Power,” a similar system is used to power their irrigation system. Differences between their proposed model and this system's proposed model is that this system has an extra power check to make sure there is always power to the system.

## **2.3 Soil Moisture Sensor System**

Soil moisture sensor systems can be implemented in a variety of ways. A simple resistive soil moisture sensor can be used to measure the resistance of the soil, which is proportional to the amount of water content in the soil. However, the resistance of soil is also dependent on temperature, thus an accompanying temperature sensor would increase the accuracy of the sensor.

Another strategy would be to use a dielectric moisture sensor. These sensors are essentially capacitors, where the soil becomes the dielectric part of a capacitor and since the dielectric constant (ratio of permittivity of a material to the permittivity of air) changes as moisture builds (as the permittivity changes), so will the capacitance of the sensor.

### **2.4 Soil Moisture Sensor Technology Comparison**

The research survey of current soil moisture sensor technology is needed to select the most efficient and feasible sensor for this project. The research details a case study of four different types of moisture sensor (Capacitive dielectric sensor, two different types of conductivity sensors, and a resistance block sensor) to compare their performance based on the amount of water (in inches) applied with ET (EvapoTranspiration) information. The ET information simply tells how much water should be used and by subtracting the rainfall from ET values will give the required amount of water that should be used. The case study was conducted both in clay soil and sand soil from May to September. Each plot is roughly 9' x 9' in size and each sensor was programmed to water daily with four start times of five minutes each, with at least one hour between cycles [7]. The results show that the best water saving sensor in clay soil is Capacitive dielectric sensor (Total of 11.30 inches while water requirement is 15.57 inches), and the best water saving sensor in sand.

Soil is the resistance block sensor (Total of 22.88 inches while water requirement is 15.57 inches). These results imply that the environmental factors impact the moisture sensor performance and accuracy in multiple different soils. Other than environmental factors, there are many tradeoffs between cost and accuracy, and also power consumption. The overview of the sensor technology comparison is shown below Table.

## 2.5 Table

| Sensor                                                    | Method                                                                                                                                                                                                                                      | Observations                                                                                                                                                               | Observations                                                                                                 |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Fiber Optics                                              | Measures the change of the intensity of the light travel exiting the probe to soil and entering back to the probe from soil.                                                                                                                | Takes into consideration weather predictions                                                                                                                               | Expensive<br>-Very new technology                                                                            |
| Tensiometers                                              | The built-in vacuum gauge inside the plastic tube measures the pressure<br>-The pressure changes by the water pulled out of the soil<br>-Measured in centimeter: Higher reading means less moisture and a lower reading means high moisture | Easy to read the data<br>-Easy to operate<br>-Fast measurement<br>-Low power consumption                                                                                   | -\$45-80 (vary by the length, 6-48" of the probe)<br>-Needs routine maintenance<br>-Cannot use during winter |
| Electrical Resistance Blocks: Granular Matrix (Watermark) | -Similar method to Gypsum Blocks                                                                                                                                                                                                            | Better Version of Gypsum Blocks<br>-High accuracy<br>-Easy to operate<br>-Low cost (\$25-\$50)<br>-Wide range of soil moisture<br>-Long duration (5-7 years)<br>-Low power | Lower accuracy compared to dielectric sensor<br>-Slow                                                        |
| Electrical Resistance Blocks: Gypsum Blocks               | Two electrodes inside the porous material, such as gypsum, measures the resistance itself<br>-The water from the soil moves into the gypsum decreases the resistance and water pulled.                                                      | -Low cost<br>-Accurate in clay soil<br>-Easy to operate<br>-Low power consumption                                                                                          | -Duration is around 1-2 years<br>-Low accuracy in sandy soils<br>-Less repeatability                         |

Fig 1: Literature Survey Table

## CHAPTER 3

### METHODOLOGY

#### 3.1 Architecture



Fig 2: Architecture of Methodology



### **3.1.1 LPC-1769 Description**

The LPC1769 are ARM Cortex-M3 based microcontrollers for embedded applications featuring a high level of integration and low power consumption. The Arm Cortex-M3 is a next generation core that offers system enhancements such as enhanced debug features and a higher level of support block integration.

The LPC1769 operates at CPU frequencies of up to 100 MHz. The LPC1769 operates at CPU frequencies of up to 120 MHz. The Arm Cortex-M3 CPU incorporates a 3-stage pipeline and uses a Harvard architecture with separate local instruction and data buses as well as a third bus for peripherals. The Arm Cortex-M3 CPU also includes an internal prefetch unit that supports speculative branching.

The peripheral complement of the LPC1769 includes up to 512 kB of flash memory, up to 64 kB of data memory, Ethernet MAC, USB Device/Host/OTG interface, 8-channel general purpose DMA controller, 4 UARTs, 2 CAN channels, 2 SSP controllers, SPI interface, 3 I2C-bus interfaces, 2-input plus 2-output I2S-bus interface, 8-channel 12-bit ADC, 10-bit DAC, motor control PWM, Quadrature Encoder interface, four general purpose timers, 6-output general purpose PWM, ultra-low power Real-Time Clock (RTC) with separate battery supply, and up to 70 general purpose I/O pins.

The LPC1769 are pin-compatible to the 100-pin LPC236x Arm7-based microcontroller series.

### 3.1.2 DIAGRAM

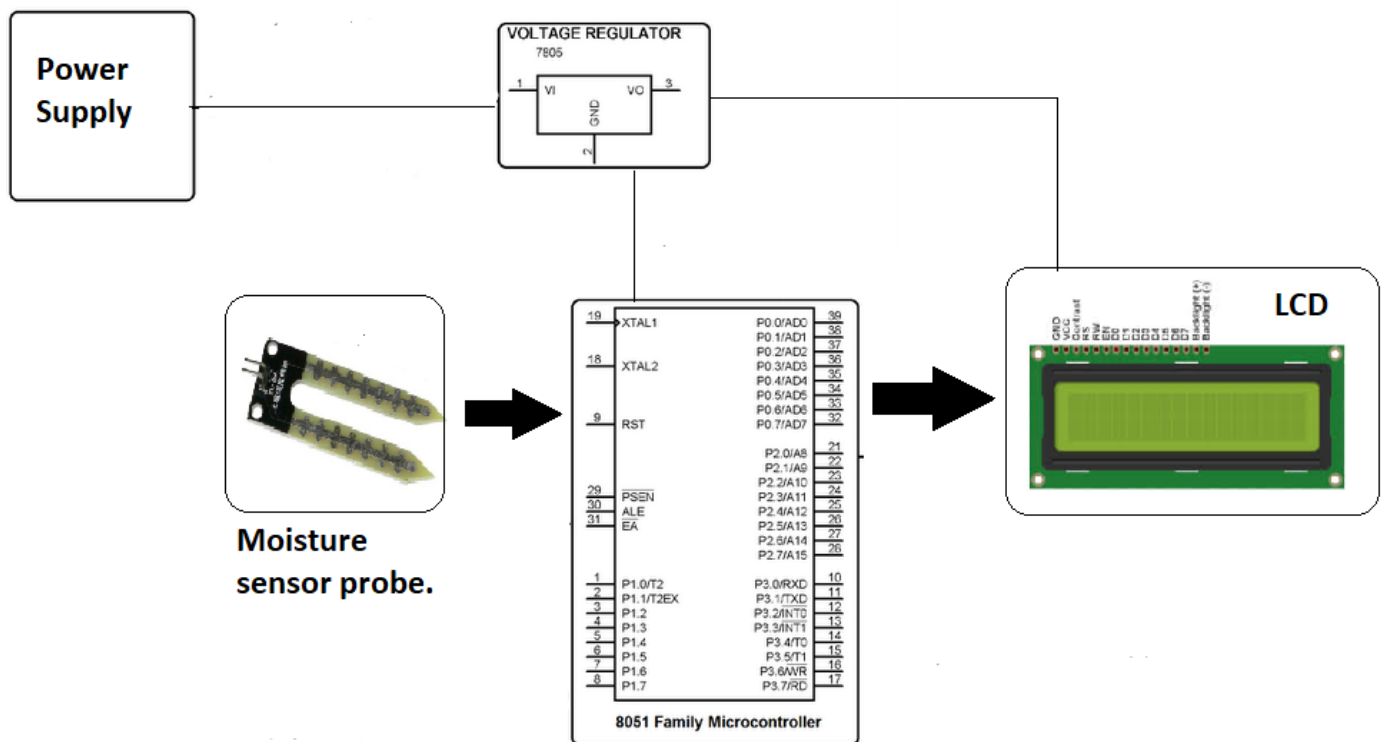


Fig 3: Diagram of Methodology

### 3.2 YL-69 Sensor

- The sensor is set up by two pieces: **the electronic board** (at the right), and **the probe** with two pads, that detects the water content (at the left).



Fig 4: Probes of YL-69 Sensor

- The sensor has a built-in potentiometer for sensitivity adjustment of the digital output (D0), a power LED and a digital output LED, as you can see in the following figure.

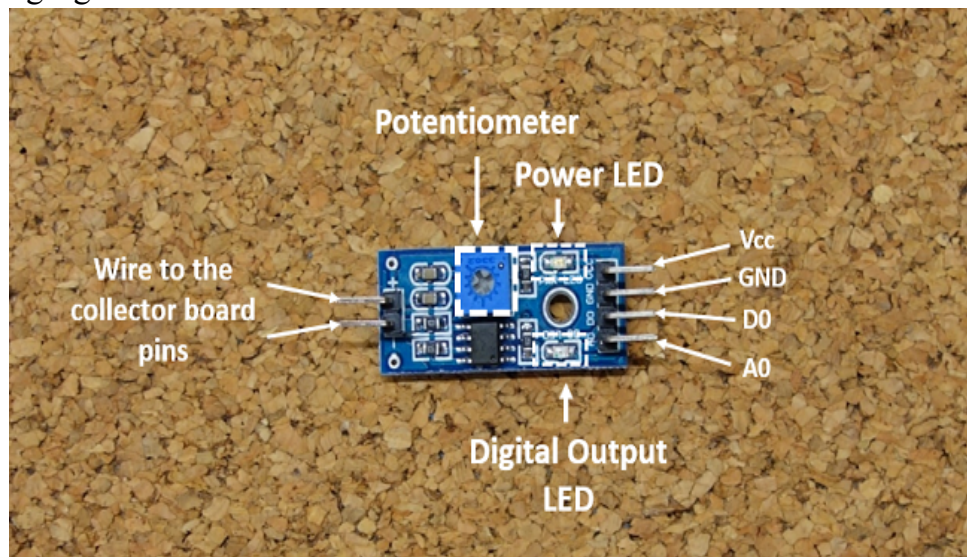


Fig 5: Electric Board of YL-69 Sensor

The output can be a digital signal (D0) LOW or HIGH, depending on the water content. If the soil humidity exceeds a predefined threshold value, the module outputs LOW, otherwise it outputs HIGH. The threshold value for the digital signal can be adjusted using the potentiometer.

### 3.2.1 Pin out of Soil moisture sensor

The pin out soil moisture sensor is given below. The connection to pic microcontroller extremely easy as follows:

- We connect the ground of the sensor to the ground pin of the power supply.
- VCC pin of sensor to 5v of power supply voltage.
- Next we connect the analog output of the sensor to analog pin 0 of the pic microcontroller.

### HOW DOES THE SENSOR WORK?

The voltage that the sensor outputs changes accordingly to the water content in the soil. When the soil is:

- **Wet:** the output voltage decreases
- **Dry:** the output voltage increases

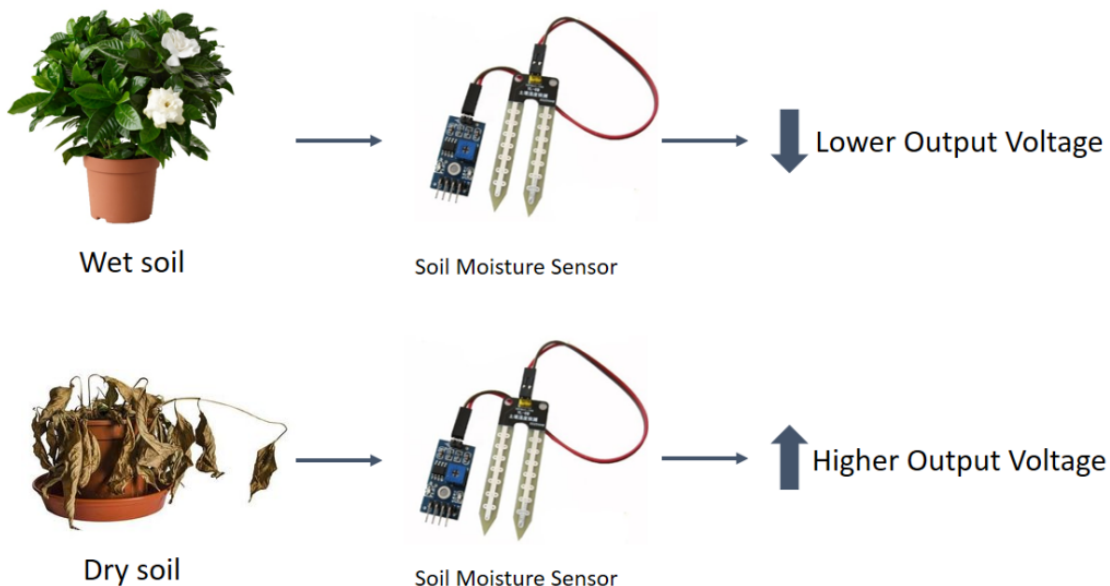


Fig 6: Sensor Working Process

### 3.2.1.1 Comparator circuit

The YL 38 comparator module is used in this project. It is a single channel op amp comparator based on L393 IC. This module just compares the output voltage of the sensing probe with a reference voltage and switches its voltage appropriately for the microcontroller to read. Circuit diagram of the YL 38 comparator module is shown below.

The output voltage of the sensing probe is connected to the inverting input of the opamp. When the moisture level is high more current passes through the sensing probe and so the voltage at the inverting pin will be higher than the reference. The reference can be set using the trim pot R2. At this condition the output of the opamp goes low and sinks the LED D1 to make it glow. When the moisture is lower than the set point, the opposite happens.

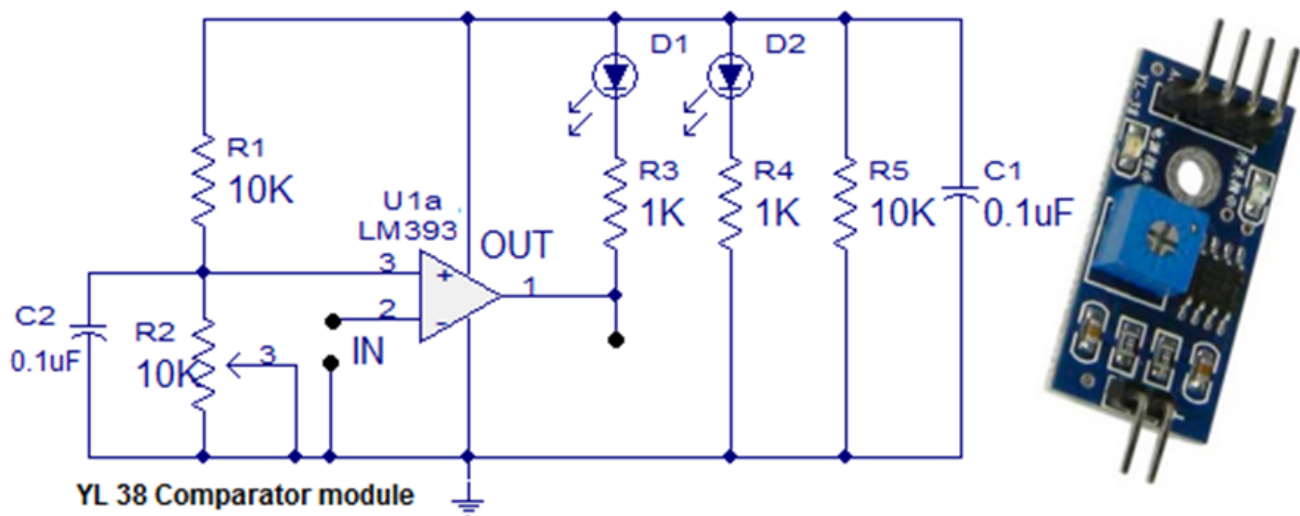


Fig 7: Comparator Circuit

### 3.3 LPC-1769 Microcontroller:

- 4 KB on chip program memory.
  - 128 bytes on chip data memory(RAM)
  - 4 reg banks.
  - 128 user defined software flags.
  - 8-bit data bus
  - 16-bit address bus
  - 16 bit timers (usually 2, but may have more, or less).
  - 3 internal and 2 external interrupts
- Microcontroller section senses the status of the comparator module and switches the irrigation motor appropriately. It also displays the status of the pump on the LCD screen. The power supply unit and the motor driver unit are incorporated in this section. Full circuit diagram of the project is given below.

### 3.3.1 MICROCONTROLLER DIAGRAM

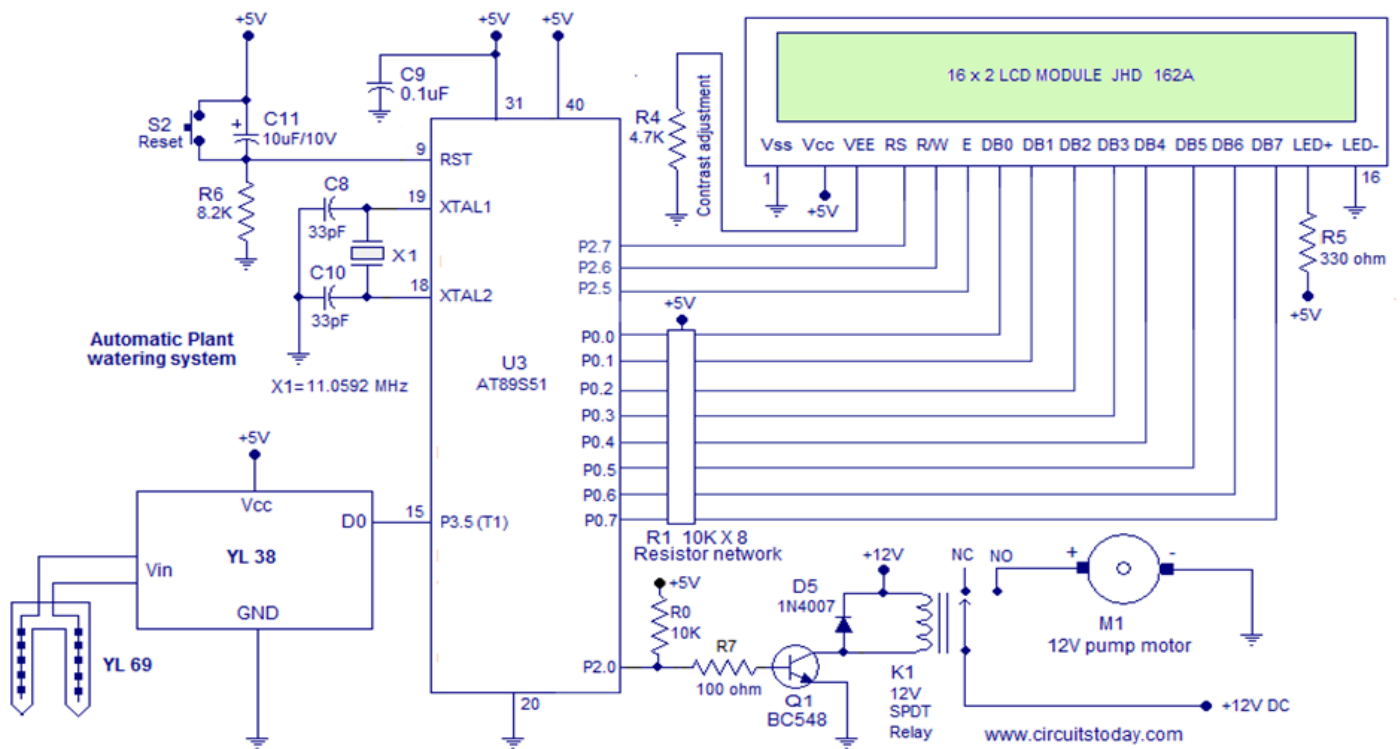


Fig 8: Microcontroller Diagram

### 3.4 Block Diagram

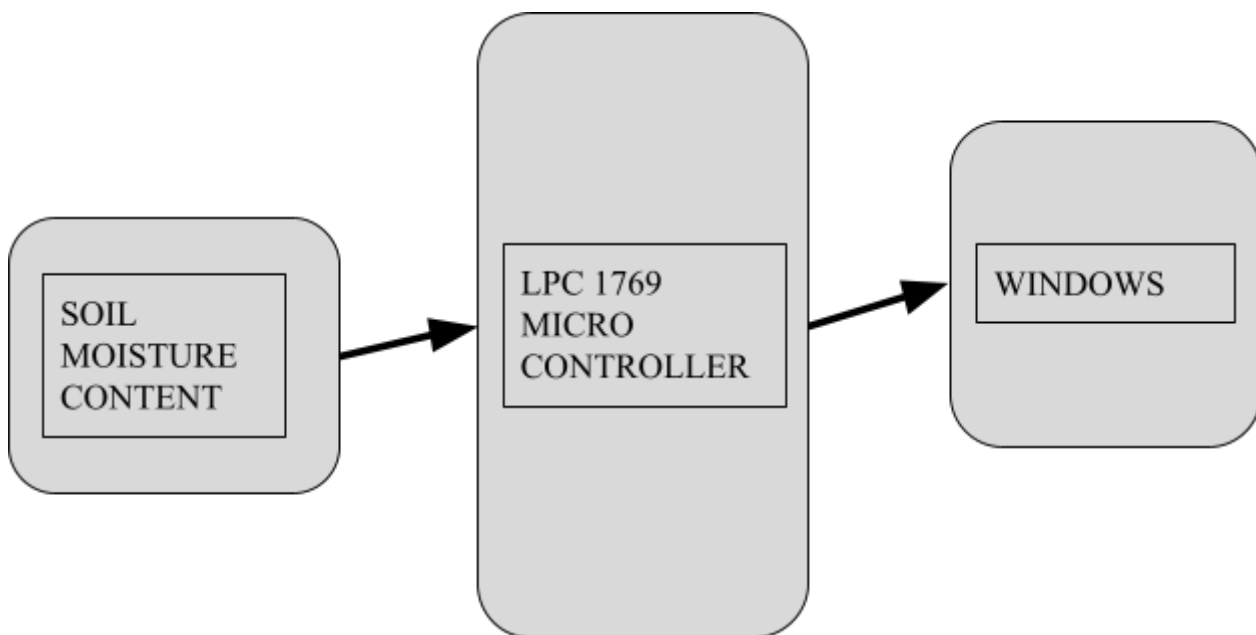


Fig 9: Block Diagram of Methodology



### **3.5 Keil MicroVision 4**

Keil MicroVision is a free software which solves many of the pain points for an embedded program developer. This software is an integrated development environment (IDE), which integrates a text editor to write programs, a compiler and it will convert your source code to hex files too.

Here is simple guide to start working with Keil uVision which can be used for

- Writing programs in C/C++ or Assembly language
- Compiling and Assembling Programs
- Debugging program
- Creating Hex and Axf file
- Testing your program without Available real Hardware (Simulator Mode)

### 3.6 Flow Chart

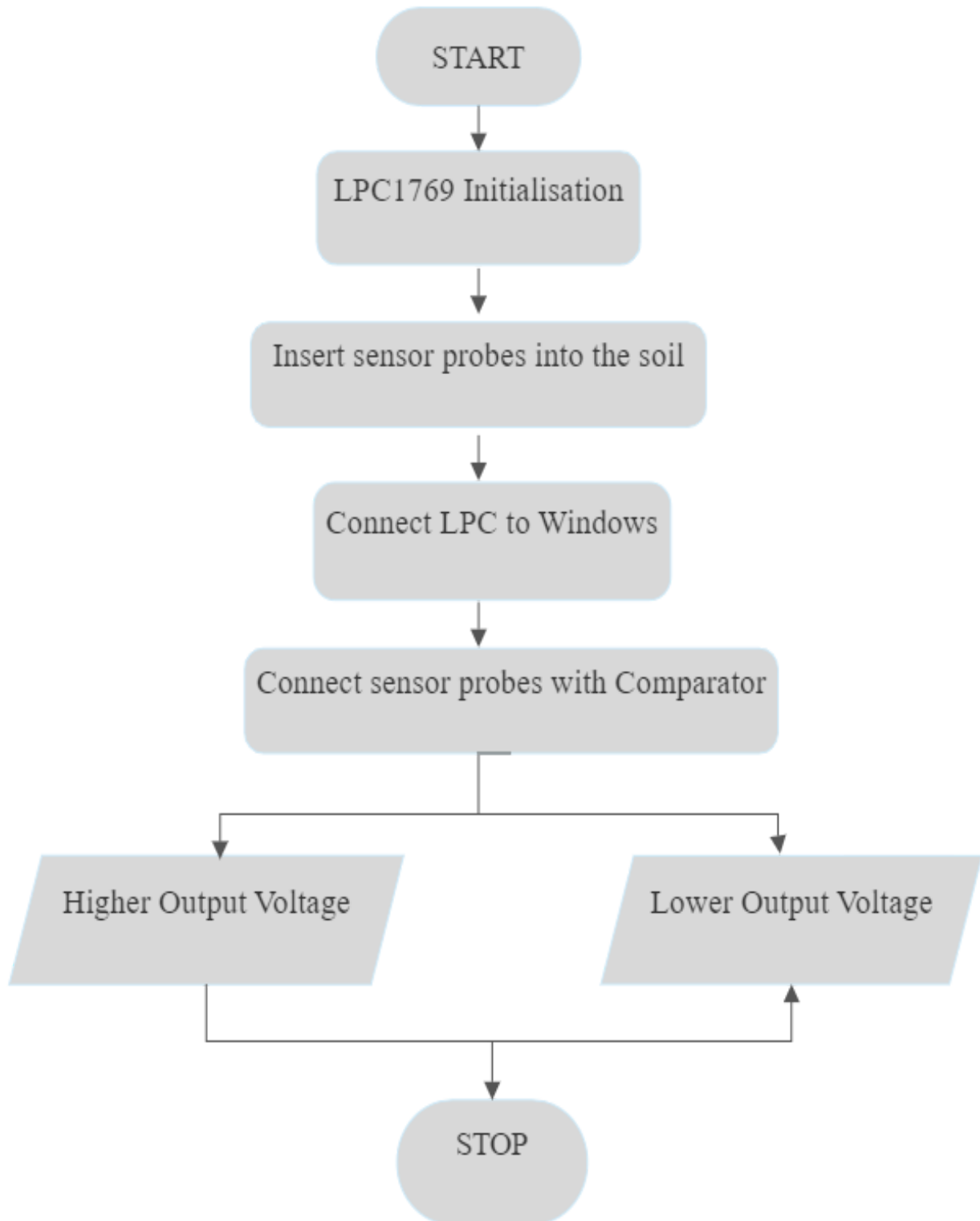


Fig 10: Flow Chart of Methodology

## CHAPTER 4

### RESULTS

#### 4.1 Output Screenshot

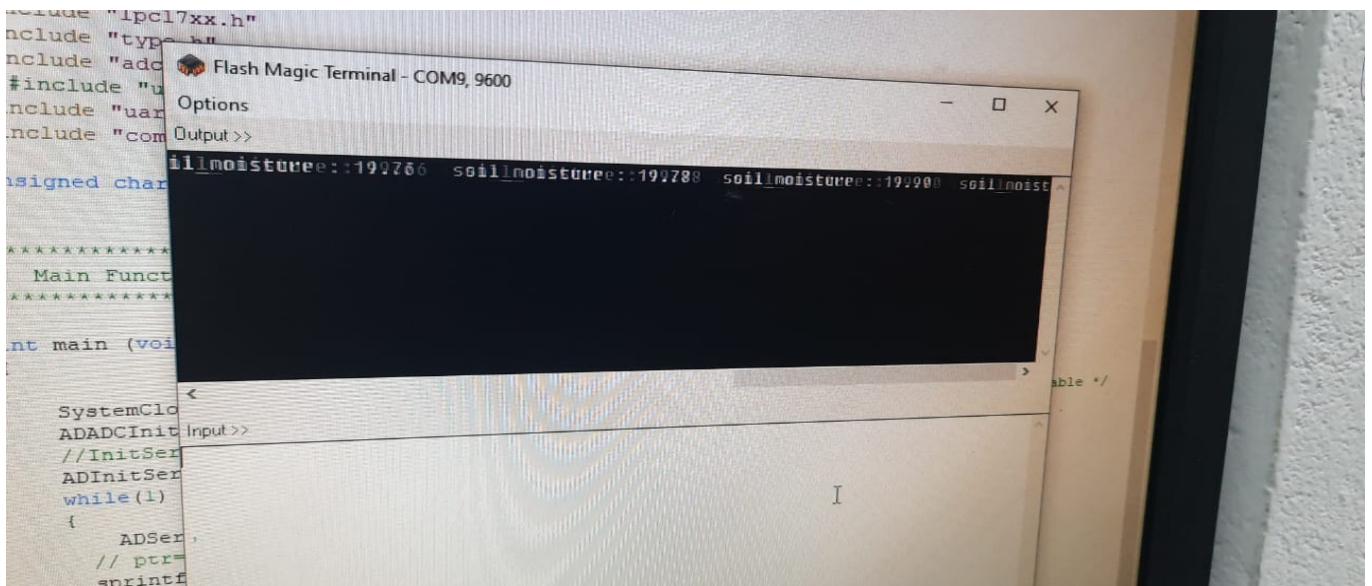


Fig 11: Output Screenshot

## **CHAPTER 5**

# **CONCLUSION AND FUTURE SCOPE**

### **5.1 Conclusion**

Sensing Moisture Content of the Soil using IoT can have a booming effect on agricultural growth. In this project we are using an attractive user interface with the most efficient way of controlling the irrigation system.

In this we have achieved the proposal to monitor the soil moisture content in a farming area and the user can control the watering system accordingly. so, the overall implementation cost is cheap and it is affordable for a common person. Considering the present situation, we have chosen the Windows system so that most people can get benefits. The design consists of a Windows System by which the user can interact and ON/OFF the pump motor accordingly. This system of irrigation is also helpful in the region where there is a scarcity of water and improves sustainability. The sensor can also be adjusted according to the needs of varieties of the crop to be irrigated.

With more advancement in the field of IoT, it can be expected in the coming years where these systems can be more efficient, much faster and less costlier. In the future, this system can be made as an intelligent system, where in the system predicts user actions, rainfall pattern, time to harvest, animal intruder in the field and communicating the information through advanced technology like IoMT can be implemented so that agricultural system can be made independent of human operation and in turn quality and huge quantity yield can be obtained.

## 5.2 Future Scope

We can include sensors like PIR motion sensor which senses the motion and detects whether a human has moved in or out of the sensor's range, DHT11 sensor can also be added which is a Humidity and Temperature sensor, which reads the temperature values from the surroundings and Raindrop sensor which is a tool for sensing rain.

We can interface LCD screens in order to display the current status of the soil moisture content levels, percentage of water utilized to water the plant, duration of time for which the water pump is ON, etc. We can also show the graphical representation of the moisture content levels in the soil.

To improve the efficiency and effectiveness of the system, the following recommendation can be put into consideration. A mobile Application can be developed in such a way that farmers can control the irrigation system using their mobile.

The idea of using the IoT for irrigation can be extended further to other activities in farming such as cattle management, fire detection and climate control. This would minimize human intervention in farming activities.

The future scope of this project is, it can also be designed to detect a particular disease on a plant and suggest the proper curative measures on it. In the same way one can predict the weather if the system is made to communicate with the nearer weather station through satellite communication.

### 5.3 References

- [1] Sukriti<sup>1</sup>, Sanyam Gupta<sup>2</sup>, Indumathy K<sup>3</sup> IoT based Smart Irrigation and Tank Monitoring System, International Journal of Innovative Research in Computer and Communication Engineering (An ISO 3297:2007 Certified Organization) Vol.4, Issue 9
- [2] IoT based Smart Irrigation Tank Monitoring System Sukriti, Sanyam Gupta, and Indumathy KB. Tech, Department of Computer Science and Technology, Vellore Institute of Technology
- [3] [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=3669704#:~:text=Soil%20moisture%20sensor%20is%20a,using%20an%20automated%20irrigation%20system](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3669704#:~:text=Soil%20moisture%20sensor%20is%20a,using%20an%20automated%20irrigation%20system).
- [4] <https://ijettjournal.org/Volume-67/Issue-3/IJETT-V67I3P210.pdf>